# Worksheet B: Dice Detectives, Base Rates, Credibility, Your Project

Name: _____ Date: _____

## Task 4: Dice detectives

in class, pairs, 10 min.

**Roles:** R = roller, D = detective. **1.** R opens the envelope and reads the card *alone*: "fair" (only a 6 counts as a hit) or "loaded" (5 and 6 count as hits). **2.** R rolls the die hidden under a cup and only says "hit" or "no hit". **3.** D makes a tally mark; stop after exactly **30** rolls. **4.** D does both analyses and decides. **5.** Only then does R reveal the card. Known in advance: one envelope in five is loaded (base rate $1:4$).

| | |
|---|---|
| Hit (tally) | |
| No hit (tally) | |

| Hits $k$ | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $p = P(X \geq k \mid \text{fair})$ | 0.90 | 0.76 | 0.58 | 0.38 | 0.22 | 0.11 | 0.051 | 0.020 | 0.007 | 0.002 | 0.0005 | 0.0001 |
| $BF_{10} = 2^k \cdot 0.8^{30-k}$ | 0.02 | 0.05 | 0.12 | 0.30 | 0.76 | 1.9 | 4.7 | 11.8 | 30 | 74 | 184 | 461 |
| $P(\text{loaded})$ at $1:4$ | 0.5 % | 1 % | 3 % | 7 % | 16 % | 32 % | 54 % | 75 % | 88 % | 95 % | 98 % | 99 % |

1. **Frequentist:** $H_0$: fair. p-value for your $k$: _____. At $\alpha = 0.05$ the die is ☐ significantly loaded ☐ not significant.
2. **Bayesian:** $BF_{10}$ = _____; prior odds $1:4$, posterior odds _____; $P(\text{loaded})$ = _____. Do the calculation once yourself, without the last row of the table.
3. Open the envelope. Who was right? Enter your pair in the two-by-two table on the board.
4. What would your Bayesian verdict be if only one die in 100 were loaded? And your test verdict?

## Task 5: Base rate: the taxi and the witness

homework

In a city, 85 % of taxis are green and 15 % are blue. After a hit-and-run accident, a witness says the taxi was blue. Tests show that under the visibility conditions of that night he identifies the colour correctly in 80 % of cases (Tversky and Kahneman 1982).

1. First, a gut estimate: how likely was the taxi blue?
2. Calculate with natural frequencies (100 accidents) and draw the tree as on the slide about the medical test.
3. Do the same calculation in odds form: prior odds, the witness's Bayes factor, posterior odds.
4. A second, independent witness with the same accuracy also says "blue". How likely is it now?
5. In another city, half of the taxis are blue. With one witness, what changes and what does not?

## Task 6: Credibility check with a search engine

in class, small groups, 12 min.

Claim (card): _____

Base rate *before* searching: of 100 claims of this kind, about _____ are true, because

| **Finding** (source, what does it say?) | **expected if true? if false?** | **Factor** | **Odds** |
|---|---|---|---|
| Start: base rate | | – | |
| | | | |
| | | | |
| | | | |
| | | | |

Rough factors: $\times 10$, $\times 3$, $\times 1$, $\times \frac{1}{3}$, $\times \frac{1}{10}$. Only multiply independent findings; ten sites copying the same press release are *one* finding. Result: $P(\text{true}) \approx$ _____. Which finding moved you the most?

## Task 7: Reading a news report

homework

**Fictitious report** (written for this exercise):

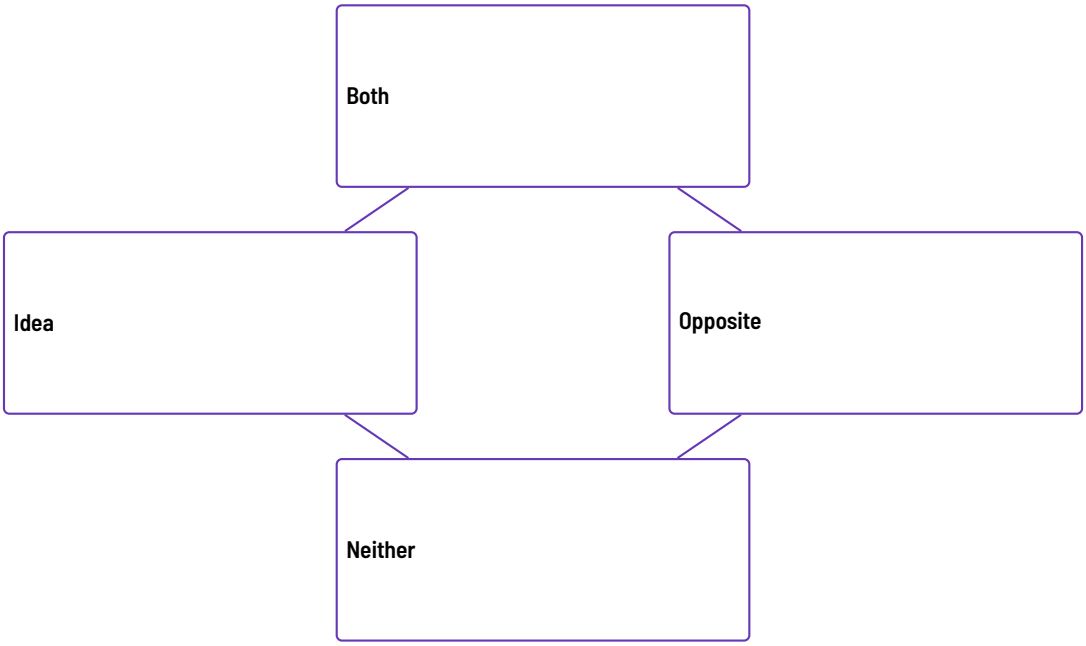
> *Magic tricks make you smarter – now proven.* People who watch a magic trick before studying remember more. This is shown by an online survey of 112 students. The researchers recorded 23 aspects of learning behaviour. On the item "I was able to concentrate well", the group that saw a magic trick came out significantly ahead ($p = 0.04$). "This scientifically proves the value of magic in the lecture hall", says the head of the study.

Answer: What was measured, and in whom? What is claimed? Is an effect size reported? How many comparisons were made, and what does that mean for the one significant result? What base rate would you assign to the claim? If you like, apply the same questions to a real article that presents a study with $p < 0.05$ as "proof".

## Task 8: Diamond and map for your idea

homework, A3, bring to Unit 7

Take your hypothesis from Unit 5 as the *idea* and fill in the diamond. At each position, note which observation it *allows* and which it *rules out*.




1. Which observation separates the idea from its opposite? Which separates the idea from the "neither" position? Roughly estimate how likely the observation is under each position and compute the Bayes factor.
2. Write down the new, sharpened statement.
3. On an A3 sheet, draw the map of your study (legend on the slide): starting village, paths, fog, dragons, bridges, dead ends, blank spots, destination. Every element carries a source or a number.

## Task 9: Analysis plan for your project

homework, basis for the design sketch in Unit 7

| | |
|---|---|
| $H_1$ (from Unit 5) | |
| Outcome variable and level of measurement | |
| Comparison: two groups, repeated measures or proportions? | |
| Smallest effect that would interest you | |
| Realistic $n$ (who, how many?) | |
| Power according to the table on the slide | |
| Analysis: test, estimate with interval, or Bayes factor with stopping rule? | |
| Base rate of your hypothesis: how surprising would it be if it were true? | |
| Conclusion: does the question fit the $n$, or will you change the question or the design? | |

### Task 10: Diary of wonder

homework

When did your sense of probability last deceive you? Write down what you expected, what happened, and the matching concept from this unit.